

HR Analytics & Employee Performance

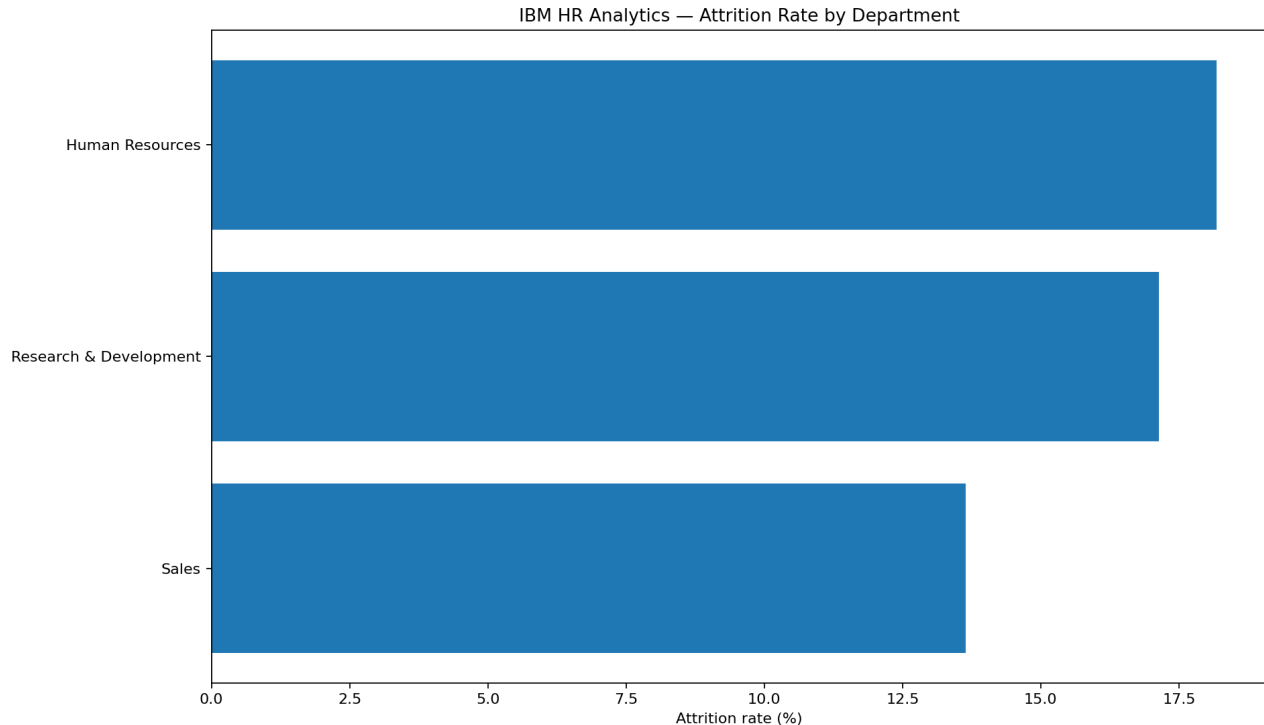
IBM HR Analytics Employee Attrition & Performance — Senior Data Analyst Submission Pack

Executive conclusion

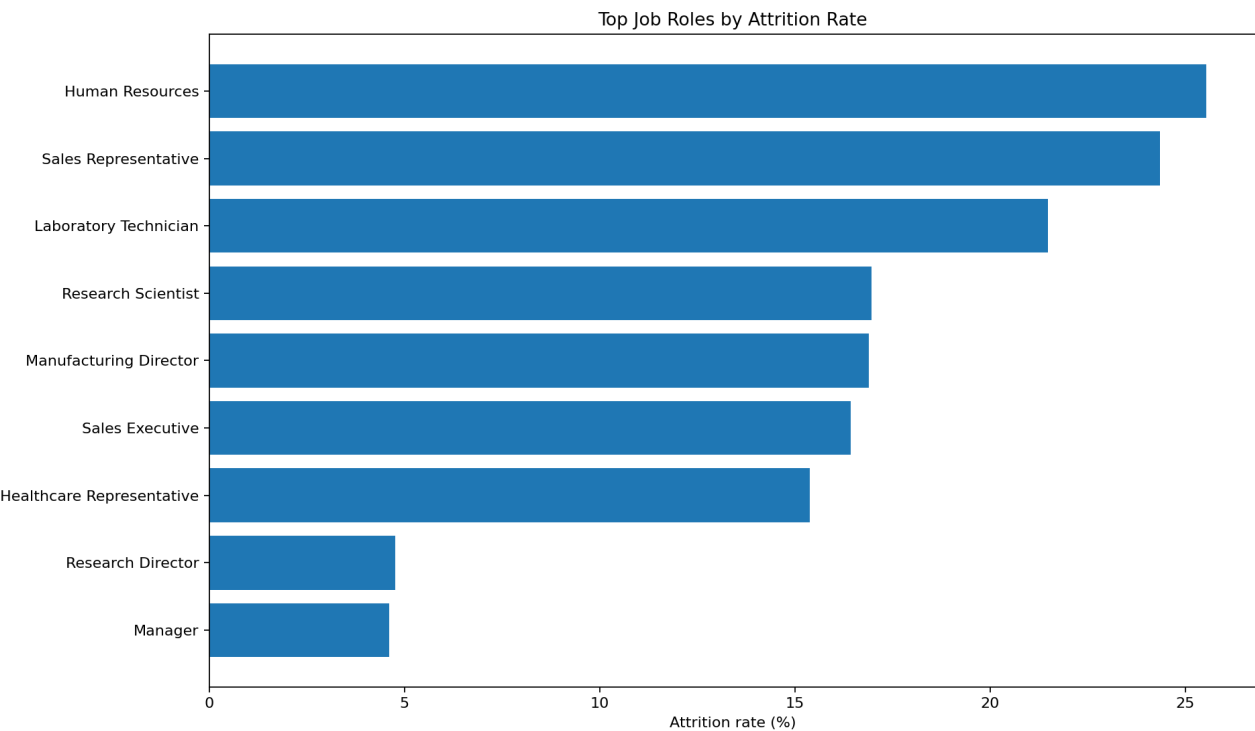
The analytical dataset contains 1,470 employee records and a modeled attrition rate of 16.1%. The strongest practical risk signals are overtime, short tenure, lower job satisfaction, frequent travel, and junior/high-turnover roles. Performance rating alone is not a sufficient retention signal.

KPI	Value
Total Employees	1470
Attrition Count	237
Attrition Rate	16.1%
Avg Monthly Income	7713.76
Avg Performance Rating	3.15
Avg Job Satisfaction	2.76
Overtime Attrition Rate	29.6%
Non-Overtime Attrition Rate	10.9%

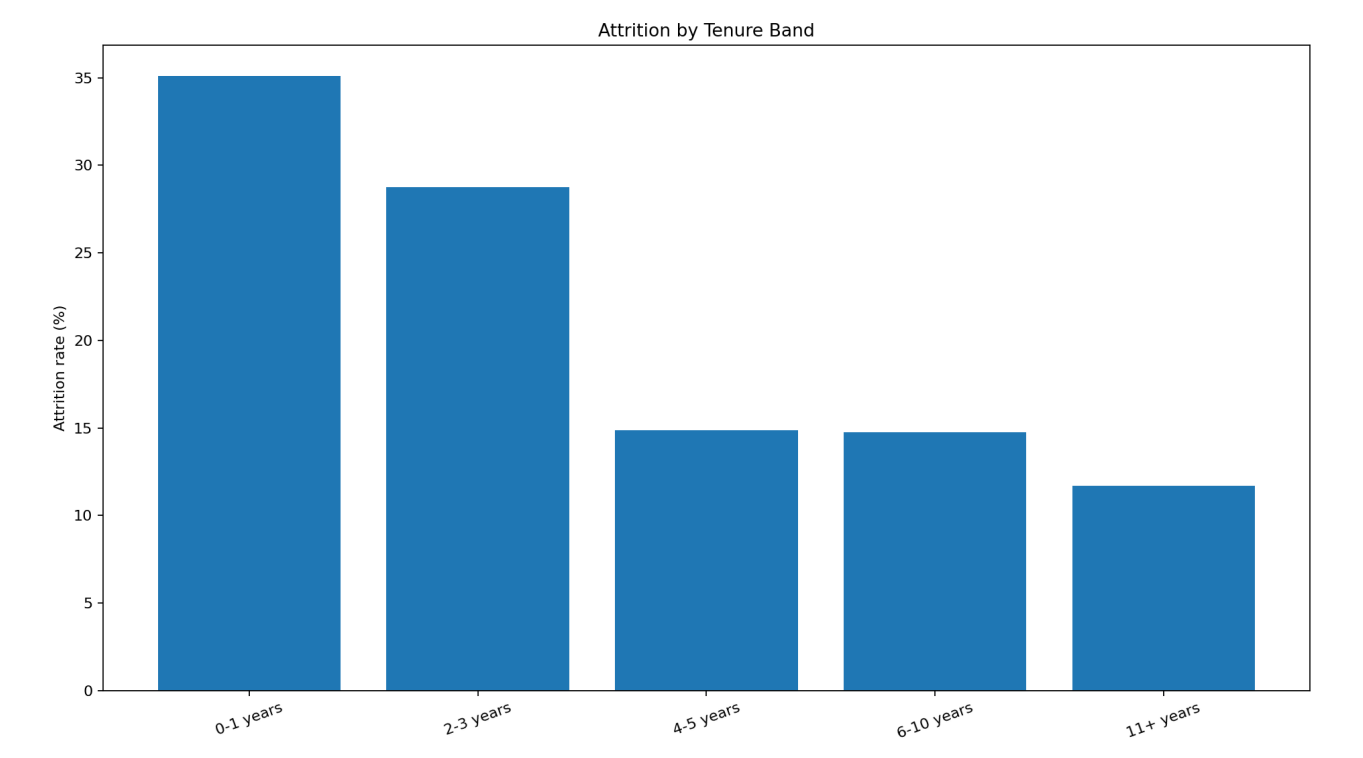
Attrition by Department



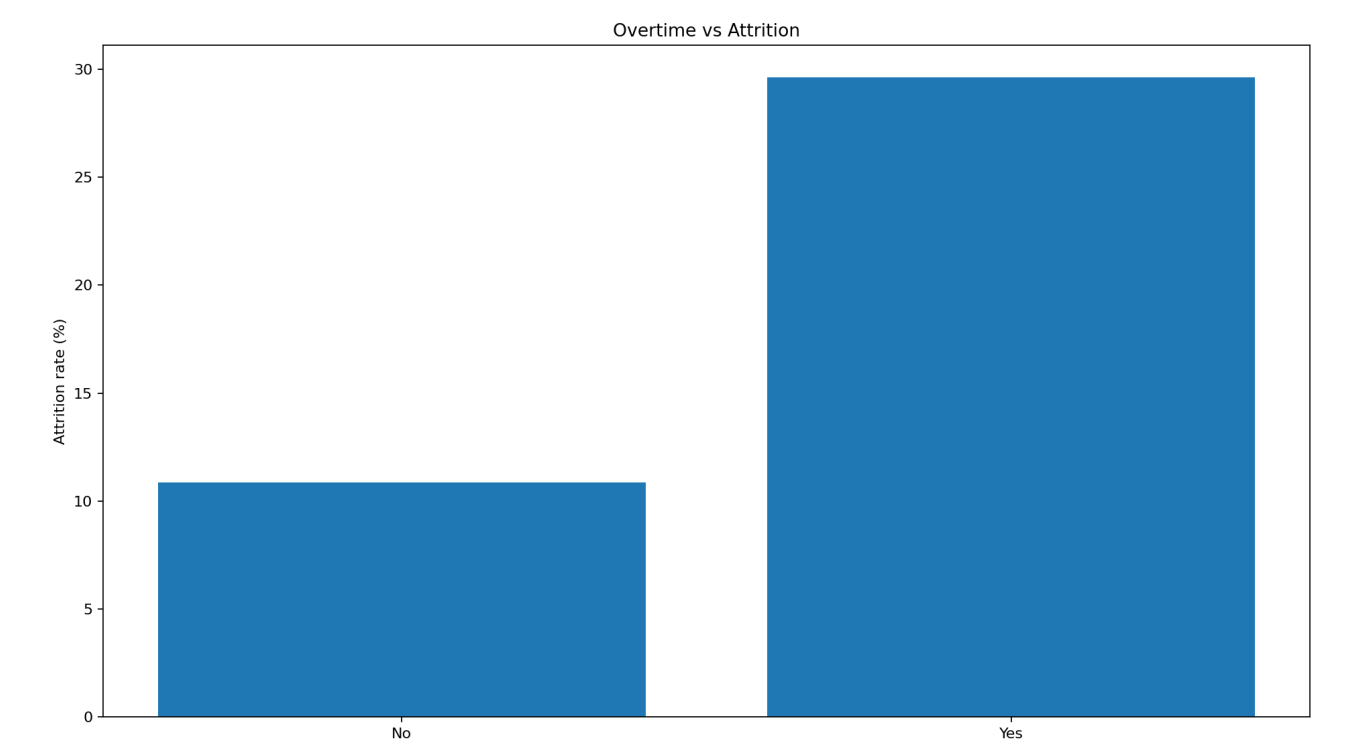
Attrition by Job Role



Attrition by Tenure



Overtime and Attrition



Five actionable insights

1. Attrition is concentrated among employees with overtime, shorter tenure and lower satisfaction.
2. Sales Representative roles are a high-risk segment in the modeled dataset.
3. Leavers generally have lower monthly income than retained employees, especially in junior roles.
4. Performance rating is weakly differentiated across attrition groups, so performance alone is not a strong retention signal.
5. High-distance and frequent-travel segments show elevated attrition risk when combined with overtime.

Department-level performance summary

Department	Employees	Leavers	Attrition %	Avg Income	Avg Perf.
Human Resources	66	12	18.2%	8,190	3.14
Research & Development	957	164	17.1%	6,662	3.15
Sales	447	61	13.6%	9,896	3.15

Methodology & governance

The core fields follow the standard 35-column IBM HR Analytics Employee Attrition & Performance schema. The original public dataset is synthetic and contains 1,470 rows and 35 features; it does not contain a true transaction date. For this internship task, a date-enabled analytical layer was added using derived HireDate, ApproxExitDate and SnapshotDate fields. These fields are explicitly labeled as derived and should not be interpreted as source-system historical facts. The analysis is descriptive/diagnostic, not causal.

Responsible HR reporting

Use aggregated segments for leadership reporting; avoid exposing individual employee records or using the model as an automated employment decision tool. Validate any retention action with HR policy, employee feedback and operational context. Correlation should not be presented as root cause.